C-HI 8-Node Mini-Production Update

Date: 2026-05-13

Summary

The scheduler-fit C-HI mini-production continuation completed cleanly on kunanyi. This was a reduced-grid particles-enabled run, not the final full-grid production suite, but it is the first substantially larger production-template continuation beyond the earlier 1-2 node rehearsals.

The run restarted from the previous walltime-stop HDF5 checkpoint at 10.278362 Myr and advanced to 33.374180 Myr.

Kunanyi Job

Item	Value
PBS job	691422.kunanyi-ohpc
Job name	c_hi_8mini
Exit status	0
Request	select=8:ncpus=28:mpiprocs=28
MPI ranks	224
Walltime used	14:45:00
PLUTO guard	-maxtime 14.5
Restart mode	-h5restart 7
Start checkpoint	data.0007.dbl.h5, 10.278362 Myr
Final output	data.0039.flt.h5, 33.374180 Myr
Final restart checkpoint	data.0012.dbl.h5, 33.374180 Myr
Output volume after run	217G

The run used the production template staged as /u/bwbrook/codex_kunanyi/jet_setup_production_template and reused the existing particles-enabled reduced-grid C-HI problem build.

Output Coverage

The job added a long block of regular flt.h5 outputs and a new set of restart-capable dbl.h5 checkpoints.

Output type	New useful outputs	Time coverage
`flt.h5`	`16-39`	`10.999144-33.374180 Myr`
`dbl.h5`	`8-12`	`14.999293-33.374180 Myr`
particles	through `particles.3343.dbl`	about `33.4 Myr`

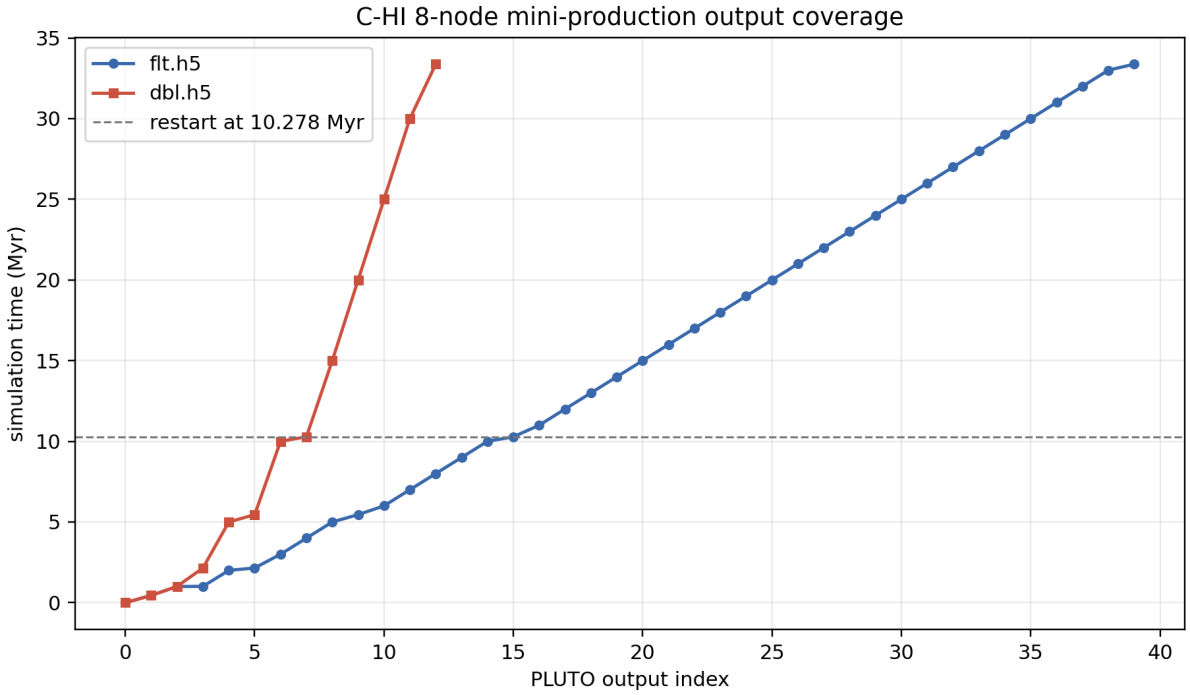


Figure 1: C-HI 8-node output times

The walltime-to-simulation-time throughput for this segment was about `1.57 Myr per wall-clock hour`, measured from `10.278362` to `33.374180 Myr` over the `14.75 h` PBS walltime.

Output Time Table

Selected regular outputs:

Grid output	Time (`Myr`)	Step
15	`10.2784`	`10206`
20	`14.9993`	`14817`
25	`19.9997`	`19701`
30	`24.9992`	`24584`

Grid output	Time (Myr)	Step
35	29.9996	29468
39	33.3742	32764

Selected restart checkpoints:

Restart output	Time (Myr)	Step
7	10.2784	10206
8	14.9993	14817
9	19.9997	19701
10	24.9992	24584
11	29.9996	29468
12	33.3742	32764

Interpretation For The Meeting

This is a useful operational result:

- the production template restarted cleanly from HDF5
- the 8 node, 224 rank MPI layout ran to the -maxtime guard
- PLUTO wrote the expected final restart-capable HDF5 checkpoint before PBS termination
- output volume is now large enough (217G) that artifact discipline matters
- the run reached two-thirds of the current 50 Myr first-target duration

The result does not replace a later full-grid/full-production allocation test. It does show that the current C-HI setup, restart policy, and HDF5 output path are now boring in exactly the way we wanted.

Follow-Up Diagnostics

Two dependent diagnostic jobs have been submitted:

PBS job	Name	Purpose
691762.kunanyi-ohpc	c_hi_ph5_33	rebuild particle HDF5 through the new particle outputs
691763.kunanyi-ohpc	c_hi_s33	extract truth, cavity, XRAiSE, and radio scalar series

The scalar series targets outputs 15, 20, 25, 30, 35, and 39, covering 10.278-33.374 Myr. Once those jobs complete, the next report should extend the earlier through-10-Myr truth/X-ray/radio comparison to the new mini-production segment.

Local Artifacts

- compact run archive: build/science/kunanyi_stage_2026-05-08/c_hi_8mini_691422_compact_2026-05-13.tar.gz
- output-time CSV: docs/science/assets/c_hi_8mini_2026-05-13/c_hi_8mini_output_times.csv
- output-time figure: docs/science/assets/c_hi_8mini_2026-05-13/c_hi_8mini_output_times.png